requirement for an increased amount of carbon dioxide should be provided for carbon dioxide fire-extinguishing systems.

4.7.4 Features of hydrogen fire

Hydrogen burns at high temperature, but generally gives off less radiant heat than propane or other hydrocarbons (e.g. only about 10% of that radiated by an equal -sized propane flame). Although the heat radiated by a hydrogen flame is also relatively low compared to hydrocarbons, it is important to take into account the differences in heats of combustion,

* Electrical equipment used in hydrogen/air mixture should be, at least, the type of "II-C" and "T-1" as the group based on the maximum experimental safe gap for flameproof enclosures and the temperature class based

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based on the maximum experimental safe gap for flameproof enclosures and the temperature class based on maximum surface temperature, respectively, according to ISO/IEC 80079-20-14).

burning rate and flame size. Hydrogen flames are colourless or nearly colourless. Both of these characteristics make it more difficult to detect a hydrogen fire. Even relatively small hydrogen fires are very difficult to extinguish. The only reliable approach to extinguish a fire is to shut off the source of hydrogen supply.

4.8 High pressure hazard

4.8.1 High pressure is a hazard common to hydrogen and other flammable gases listed in the Code. To prevent overpressure, the Code requires various measures such as pressure

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the Code. To prevent overpressure, the Code requires various measures such as pressure control and pressure design. Specifically, paragraph 8.2, in regard to the provision of pressure control of cargo tanks, requires fittings of pressure relief valves to the cargo tanks. Furthermore, paragraph 7.1.1 requires temperature control by the use of mechanical refrigeration and/or design to withstand possible increases of tempera ture and pressure. In addition, paragraph 15.2 specifies the filling limit of cargo tanks taking into account cargo volume increase by its thermal expansion. These requirements are applicable for hydrogen and no special requirement is considered necessary in this regard.

4.8.2 Boil-off may be a more significant issue for hydrogen than for LNG in particular when insulation properties have deteriorated. Means of handling boil -off gas should be carefully considered taking into account the following issues:

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insulation properties have deteriorated. Means of handling boil -off gas should be carefully considered taking into account the following issues:

.1 Re-liquefaction of hydrogen involves very specific and costly equipment. Cargo cooling in order to avoid boil-off shows the same kind of issues; and

.2 Notwithstanding the provision in paragraph 7.4.1 of the Code, thermal oxidation of hydrogen may be permitted in accordance with paragraph 1.3 of the Code.

4.8.3 The special requirements in these aspects are considered necessary.

4.9 Health hazard

4.9.1 Human safety concern under low temperature

With regard to the influences of cold hydrogen on persons' bodies, suitable protective equipment is effective. In this aspect, paragraph 14.1 of the Code requires suitable protective equipment taking into account the character of the products, therefore, no special requirement is considered necessary.

4.9.2 Static electricity

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equipment taking into account the character of the products, therefore, no special requirement is considered necessary.

4.9.2 Static electricity

Hydrogen ignition energy is very low and hydrogen can be easily ignitable by static electricity and due consideration should be given to this issue, in accordance with the requirement in the Code on suitable protective equipment.

4.9.3 Oxygen depletion and asphyxiation

Leakage of hydrogen may cause low level of oxygen and associated asphyxiation.

4.10 Wide range of flammable limits

4.10.1 Extinguishing hydrogen fire

4.10.1.1 As mentioned in paragraph 4.6, for flammable products the Code already requires elimination of sources of ignition, including use of electrical installations of appropriate types in order to minimize the risk of fire and explosion. No special requ irement is considered necessary with regard to ignitability of hydrogen.

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necessary with regard to ignitability of hydrogen.

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4.10.1.2 Furthermore, with regard to the wide range of flammable limits of hydrogen, the increased quantities of carbon dioxide as a fire -extinguishing medium should be specified as mentioned in paragraph 4.7. No additional special requirement is considered to be necessary with regard to the wide range of flammable limits of hydrogen.

4.10.2 Disposal of cold hydrogen gas

The wide flammability range makes disposal of cold hydrogen gas a major hazard. Cold plumes of released hydrogen may impede adequate dilution of hydrogen down to below 4% and may lead to flash-back to the vent from distant ignition sources outside safety -controlled areas. The low ignition energy and wide flammable range may present significant challenges.

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areas. The low ignition energy and wide flammable range may present significant challenges.
4.11 Prevention of dangerous purging operation
4.11.1 During cargo operations for maintenance, pipes and tanks should be purged with an inert gas or inert gases as illustrated in the figure below. For safety, due consideration should be given to temperature and boiling points of the inert gases. Residual pockets of hydrogen or the purge gas will remain in the enclosure if the purging rate, duration, or extent of mixing is too low. Therefore, reliable gas concentration measurements should be obtained at a number of different locations within the system for suitable purges. Temperature should also be measured at a number of locations. Oxidizing agents may exist in a hydrogen containing equipment, specifically: air, cold box atmospheres containing air diluted with nitrogen, or oxygen-enriched air that can be condensed on process pipe work within the cold box in special circumstances.

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oxygen-enriched air that can be condensed on process pipe work within the cold box in special circumstances.
4.11.2 There are special measures that may need to be put in place in order to mitigate the hazards, e.g. air should be eliminated by nitrogen purge prior to introduction of hydrogen into cargo piping or processing equipment. Nitrogen should then be eliminated by hydrogen purge, where there is a possibility of its solidification in the subsequent process.
Figure 1
References in part A
1 ISO/TR 15916:2015, Basic consideration for the safety of hydrogen systems (ISO)
2 American Institute of Aeronautics and Astronautics, "Safety Standard for Hydrogen and Hydrogen Systems (Guide to Safety of Hydrogen and Hydrogen Systems)", 2005 (AIAA)
RESOLUTION MSC.565(108) (adopted on 24 May 2024)
REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE OF LIQUEFIED HYDROGEN IN BULK

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RESOLUTION MSC.570(109) (adopted on 6 December 2024)
PERFORMANCE STANDARDS FOR A UNIVERSAL SHIPBORNE AUTOMATIC IDENTIFICATION SYSTEM (AIS)
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ANNEX 23
RESOLUTION MSC.570(109)
(adopted on 6 December 2024)
PERFORMANCE STANDARDS FOR A UNIVERSAL SHIPBORNE AUTOMATIC IDENTIFICATION SYSTEM (AIS)
THE MARITIME SAFETY COMMITTEE,
RECALLING Article 28(b) of the Convention on the International Maritime Organization

concerning the functions of the Committee,
RECALLING ALSO resolution A.886(21), by which the Assembly resolved that the functions of adopting performance standards and technical specifications, as well as amendments thereto, shall be performed by the Maritime Safety Committee on behalf of the Organization,
RECALLING FURTHER resolution MSC.74(69), annex 3, by which the Committee adopted the Recommendation on performance standards for a universal automatic identification system
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SOURCE: MSC.570(109).pdf
URL:
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the Recommendation on performance standards for a universal automatic identification system (AIS) to improve the safety of navigation,
TAKING INTO ACCOUNT resolution A.1192(33), by which the Assembly urged Member States and all relevant stakeholders to promote actions to prevent illegal operations in the maritime sector by the "dark fleet" or "shadow fleet",
RECOGNIZING the need for measures to prevent unauthorized entry or tampering of the ship's identity information in shipborne automatic identification systems (AIS),
HAVING CONSIDERED, at its 109th session, the recommendation made by the Sub-Committee on Navigation, Communications and Search and Rescue at its eleventh session,
1 ADOPTS revised Performance standards for a universal shipborne automatic identification system (AIS), set out in the annex to the present resolution;
2 RECOMMENDS that Governments ensure that AIS equipment conforms to performance standards not inferior to those specified in:

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2 RECOMMENDS that Governments ensure that AIS equipment conforms to performance standards not inferior to those specified in:
.1 the present resolution if the equipment is installed on:
.1 new ships for which the building contract is placed on or after 1 January 2029, or in the absence of the contract, the keel of which is laid or which are at a similar stage of construction on or after 1 January 2029; or
.2 ships other than those ships prescribed in sub-paragraph .1 above, all installations of the specified type, having a contractual delivery date on or after 1 January 2029, or in the absence of a contractual delivery date to the ship, actually delivered to the ship on or after 1 January 2029; or
.2 annex 3 to resolution MSC.74(69) if the equipment is installed on ships other